

Rigorous Critique and Defense of the Singular-Coefficient Alpha-Axis Model

Executive summary

The proposed model actually contains **two different ideas** that must be kept distinct if the mathematics is to remain rigorous. The first is a **split-complex / hyperbolic algebra** with a unit j satisfying $j^2 = -1$, whose norm is $x^2 - \alpha^2$ and whose zero-divisor set is the pair of null lines $x = \pm\alpha$. That structure is standard and internally consistent, but it is **not** a theory of division by zero. The second is a **formal singular-label mechanism** in which expressions like $a/0$ are represented by a tagged value such as $a\#$. That second idea can also be made consistent, but only if $a\#$ is treated as a **new formal object** produced by an external operator D_0 , not as an actual multiplicative inverse of 0. Existing algebraic alternatives in the literature that totalize division by zero go in a similar direction: wheels modify ring laws so that division is always defined, while meadows/common meadows keep commutative-ring structure but totalize inverse/division by setting either $0^{-1} = 0$ or $x/0 = \perp$. ¹

A rigorous critique is immediate: **under ordinary ring axioms there is no element K with $0 \cdot K = 1$** . This is not a philosophical objection; it is a theorem. Therefore any model that wants $1/0$ to behave like an honest number must give up at least one familiar law, typically full distributivity, zero absorption, or the very idea that division is inverse multiplication. The strongest defense of the user's idea is therefore **not** "we have solved division by zero inside the real numbers," but rather: "we have built a symbolic-algebraic extension in which division-by-zero events are preserved as labeled singular states and, optionally, embedded geometrically as boundary strata." That is a much more defensible claim.

On the geometric side, the split-complex plane gives a clean picture: the inverse of $x + j\alpha$ fails exactly on the null set $x^2 - \alpha^2 = 0$. In that sense, a "singularity" becomes a **boundary locus** rather than a single mysterious point. Extending this idea to a 5D metric

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 + d\alpha^2$$

does yield a modified gamma factor

$$\gamma_{5D} = \frac{1}{\sqrt{1 - \frac{v^2 - u^2}{c^2}}}, \quad u = \frac{d\alpha}{dt},$$

but that is **not** a consequence of standard special relativity; it is a consequence of changing spacetime signature to one with **two timelike directions**. Standard Lorentzian causality theory assumes one timelike direction; two-time models require extra structure to avoid ghosts and causality pathologies. Bars' two-time physics is a well-known example: it does not merely add a second time coordinate, it adds a gauge symmetry precisely to remove unphysical degrees of freedom. ²

The best formalism therefore depends on the goal. For **simulation and symbolic computation**, the strongest recommendation is a **formal singular-label algebra** $D_0(a) = a\#$, optionally graded by singular order. For **clean geometry**, split-complex numbers are the right language. For **fully totaled arithmetic**, wheels and common meadows are the relevant established frameworks. For **physics**, the 5D / alpha-axis model is an interesting speculative kinematics, but it is not standard SR and should be presented as such.

3

Formal variants under consideration

The central question is not whether one can write the symbol $a\#$, but what algebraic rules that symbol obeys. The variants below differ exactly on that point.

The **split-complex variant** is the standard algebra

$$\mathbb{S}_j = \mathbb{R}[j]/(j^2 - 1), \quad z = x + j\alpha.$$

It is commutative and associative, has norm

$$N(z) = z\bar{z} = (x + j\alpha)(x - j\alpha) = x^2 - \alpha^2,$$

and has zero divisors because $(1 + j)(1 - j) = 0$. It also has idempotent projectors

$$e_{\pm} = \frac{1 \pm j}{2}, \quad e_{\pm}^2 = e_{\pm}, \quad e_+e_- = 0.$$

Geometrically, it models a $1 + 1$ Minkowski plane; algebraically, it is a perfectly ordinary commutative algebra over $\mathbb{R}$. But it does **not** make j into $1/0$. In fact, the same source emphasizes that $j^2 = 1$ and that j itself is not the square of another split-complex number, which already shows that “hyperbolic unit” and “singularity label” are different notions. 4

The **formal singular-label operator** is best written as an external map

$$D_0 : R \rightarrow M, \quad D_0(a) = a\#,$$

where R is the ordinary coefficient ring (say $\mathbb{R}$) and M is a new module or graded algebra that stores singular data. The crucial point is that $D_0(a)$ is **not** defined by the ring equation $0 \cdot D_0(a) = a$. It is a tagging operation: “the expression $a/0$ occurred.” This preserves numerator information without collapsing ordinary algebra.

A convenient realization is the **graded singular algebra**

$$R[\sigma] = \left\{ \sum_{k=0}^n a_k \sigma^k : a_k \in R \right\},$$

with $\deg \sigma = 1$. Then one can identify

$$D_0(a) = a\sigma,$$

and interpret $\sigma^2, \sigma^3, \dots$ as higher singular orders. If one wants to model square-root singularities such as $\gamma \sim \Delta^{-1/2}$, one can pass from integer grading to a semigroup grading such as $R[\sigma^{1/2}]$ or $R[\sigma^\Lambda]$ with $\Lambda \subset \mathbb{Q}_{\geq 0}$. This is not a standard named theory; it is a mathematically routine construction using polynomial or semigroup algebras, deployed here with a singular-tag interpretation.

The **wheel variant** is an established alternative in universal algebra. Carlström shows that commutative rings can be extended to **wheels**, structures in which addition, multiplication, and division are always available, but ring distributivity is modified and $0x = 0$ no longer holds in general. One of the characteristic wheel identities is

$$(x + y)z + 0z = xz + yz,$$

rather than the ring law $(x + y)z = xz + yz$. Carlström also notes that in wheels one generally has $x/x = 1 + 0x/x$, not $x/x = 1$. This is a serious algebraic option, but it is not “ordinary arithmetic plus 0^{-1} .” ⁵

The **meadow variant** is another established alternative. In an ordinary meadow, one keeps a commutative ring with a total inverse operation and imposes axioms implying

$$0^{-1} = 0.$$

Bergstra, Hirshfeld, and Tucker describe a meadow as a commutative ring with a total inverse operator satisfying the reflection and restricted inverse laws. In a **common meadow**, one instead adjoins an error value $\perp$ with

$$x/0 = \perp.$$

This is closer to modern symbolic-computation practice because a singular event propagates as an error element rather than pretending to be an ordinary number. ⁶

The **near-ring variant** weakens ring laws in a different direction. Classical near-rings drop one distributive law and allow the additive group to be nonabelian. Ligh’s paper defines a near-ring with addition forming a group, multiplication a semigroup, and only one distributive law. He also points out that the set of all maps on an additive group, with pointwise addition and composition, is a natural near-ring example. That makes near-rings relevant here because they provide a mathematically legitimate place where some familiar zero-absorption identities fail. But near-rings are poor candidates for “number systems” unless one is deliberately moving away from numerical intuition. ⁷

The **5D metric variant** is not an algebraic extension at all. It is a kinematic proposal: add an α -coordinate and define

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 + d\alpha^2.$$

From the proper-time relation $ds^2 = c^2 d\tau^2$, one derives a modified dilation factor

$$\gamma_{5D} = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v_x^2 + v_y^2 + v_z^2 - v_\alpha^2}{c^2}}}.$$

This is mathematically self-consistent as a pseudo-Riemannian ansatz, but physically it implies two timelike directions, so it is outside ordinary Lorentzian spacetime theory. ⁸

The table below synthesizes those variants.

The table is a synthesis of the split-complex source, Carlström's wheel paper, the meadow/common-meadow papers, and the near-ring literature, together with explicit derivations given later in this report.

9

Variant	Carrier	Core law	$0 \cdot x$ rule	Distributivity	Does 0^{-1} exist?	Best interpretation
Split-complex	$\mathbb{R}[j]/(j^2 - 1)$	$j^2 = 1$	Standard: $0x = 0$	Full ring distributivity	No	Geometry of null boundary
Formal singular label	$R[\sigma]$ or $R[\sigma^\Lambda]$	$D_0(a) = a\sigma$	Standard internally; tag created externally	Full ring distributivity internally	No	Symbolic singular bookkeeping
Wheel	Wheel algebra H	Total $+, \cdot, /$	Not generally $0x = 0$	Modified wheel laws	Only in wheel sense	Algebra with total division
Meadow	Commutative ring + total inverse	$0^{-1} = 0$	Standard	Full ring laws	Yes, but totalized as 0	Equational totalization
Common meadow	Ring/field + $\perp$	$x/0 = \perp$	Standard for ring part	Full ring laws plus error behavior	No ordinary inverse; error element instead	Safe total arithmetic
Near-ring	One-sided distributive algebra	one distributive law only	Depends on sidedness	Only left or right	Not generally	Weak algebraic playground
5D metric	pseudo-Riemannian manifold	extra α kinematics	Not an algebraic question	Not applicable	Not applicable	Speculative modified spacetime

Impossibility results and consistent constructions

The foundational impossibility result is short and decisive.

Theorem. In any ring R with additive inverses and distributivity, there is no $K \in R$ such that $0K = 1$.

Proof.

$$0K = (0 + 0)K = 0K + 0K.$$

Add the additive inverse of $0K$ to both sides:

$$0 = 0K.$$

Hence $0K$ must equal 0 for every K , so $0K = 1$ is impossible unless $0 = 1$, in which case the ring is trivial. $\square$

This theorem is the rigorous reason the phrase “let $1/0$ be a number” cannot work inside ordinary ring arithmetic. No amount of notation changes that fact.

A second impossibility result concerns the conflation of the hyperbolic unit j with the singular symbol $\#$. If j is a split-complex unit, then $j^2 = 1$ is perfectly consistent because j lives in a commutative algebra with degree 0. But if $\#$ is meant to encode “one division by zero,” then the natural graded semantics assigns

$$\deg(\#) = 1, \quad \deg(\#^2) = 2, \quad \deg(1) = 0.$$

Therefore any **graded** homomorphism that identifies $\#$ with j would force a degree-1 element to square to degree 0:

$$\deg(\#^2) = 2, \quad \deg(j^2) = \deg(1) = 0,$$

which is impossible in the graded category. So the correct conclusion is not “ $j^2 = 1$ is wrong,” but rather “ j and $\#$ are different symbols with different semantics.” This resolves the user’s legitimate objection.

A consistent **formal singular-coefficient model** can be built as follows. Let

$$S = R[\sigma]$$

be the polynomial ring in one formal indeterminate σ , and let

$$D_0(a) = a\sigma.$$

Define also coefficient extraction

$$\text{coef}_1 \left(\sum_{k=0}^n a_k \sigma^k \right) = a_1.$$

Then

$$\text{coef}_1(D_0(a)) = a.$$

This gives exactly the bookkeeping behavior the user wants: the numerator is preserved. The operations are ordinary polynomial operations:

$$(a\sigma) + (b\sigma) = (a + b)\sigma, \quad (a\sigma)(b\sigma) = ab\sigma^2.$$

The model is consistent because it is just a polynomial ring. It does **not** assert that σ is the multiplicative inverse of 0; rather, it asserts that σ is a formal singular label.

If one wants $0/0$ to remain distinguishable from 0, one can extend the model with a special bottom element $\perp_0$ and define

$$D_0(0) = \perp_0.$$

Then arithmetic involving $\perp_0$ can be made partial or error-propagating. This sacrifices pure-ring structure, but that is exactly what all serious totalized division theories do in one way or another. Common meadows make the same move with a global error element $\perp$. ¹⁰

A more refined version is useful for Lorentz-type singularities. Since

$$\gamma \sim \Delta^{-1/2},$$

the singular order is “half-order,” not first order. One can therefore work in a semigroup algebra with fractional grading:

$$S_{\frac{1}{2}} = R[\sigma^{1/2}], \quad D_{\sqrt{0}}(a) = a \sigma^{1/2}.$$

Then

$$(a \sigma^{1/2})(b \sigma^{1/2}) = ab \sigma.$$

This cleanly separates $1/\sqrt{0}$ from $1/0$.

A consistent **near-ring construction** shows how one can make $0 \cdot K \neq 0$ if one is willing to give up full ring laws. Let G be any abelian group under $+$, and define multiplication by **right projection**

$$a \star b = b.$$

Then $(G, +, \star)$ is a left near-ring:

$$a \star (b + c) = b + c = (a \star b) + (a \star c),$$

and

$$(a \star b) \star c = b \star c = c = a \star c = a \star (b \star c),$$

so multiplication is associative. But now

$$0 \star K = K,$$

so $0 \star K \neq 0$ whenever $K \neq 0$. If we happen to designate some nonzero element by the symbol 1, then $0 \star 1 = 1$. This is a rigorous toy example showing that once ordinary ring laws are abandoned, equations like $0 \cdot K = 1$ become possible in some algebraic sense. The price is steep: there is no multiplicative identity in the ordinary sense, no field behavior, and no useful inversion theory. It is a demonstration of **logical possibility after axiom change**, not a satisfactory arithmetic of division by zero.

A more natural near-ring example comes from mappings. Ligh notes that if one takes all maps $f : G \rightarrow G$ on an additive group G , with pointwise addition and multiplication by composition, one obtains a near-ring. If one uses the left-near-ring convention

$$f \star g = g \circ f,$$

then

$$0 \star f = f \circ 0,$$

which can be a nonzero constant map if $f(0) \neq 0$. Again, the lesson is the same: relaxing ring axioms can produce non-absorbing zero behavior, but only in a weaker algebraic universe. ¹¹

Wheels provide the strongest existing “defense” for the ambition to define division by zero as an actual operation. Carlström’s theory proves that one can extend commutative rings to wheel structures where division is always available, but the algebra must change. In particular, full distributivity is replaced by weaker laws involving zero-terms, and $0x = 0$ fails in general. This means wheels are a mathematically serious example of the broader philosophical point behind the user’s proposal: **division by zero is possible only after changing the ambient algebra.** ⁵

Meadows defend a different point: not every totalization has to generate a whole family $a\#$. Meadows totalize by sending 0^{-1} to 0, while common meadows totalize by sending $x/0$ to a distinguished error element $\perp$. Therefore meadows and common meadows show that the user’s proposal is **not forced** by the mere desire to totalize division. It is one particular design choice among several. ¹²

Geometric embedding and the extended metric

The cleanest geometric part of the proposal is the split-complex plane. Let

$$z = x + j\alpha, \quad j^2 = 1.$$

Then

$$\bar{z} = x - j\alpha, \quad N(z) = z\bar{z} = x^2 - \alpha^2.$$

Whenever $N(z) \neq 0$, $=$

$$z^{-1} = \frac{\bar{z}}{N(z)} = \frac{x - j\alpha}{x^2 - \alpha^2}.$$

Therefore the inverse fails exactly when

$$x^2 - \alpha^2 = 0,$$

that is, on the two null lines

$$\alpha = \pm x.$$

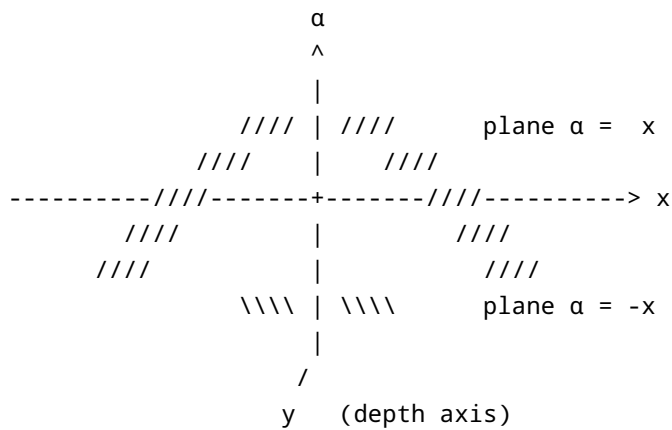
Because $(1 + j)(1 - j) = 0$, these are zero-divisor directions, not ordinary points. In this precise sense, the split-complex algebra turns “a singular failure of inversion” into a **geometric boundary set.** ¹³

That geometric picture is easiest to visualize in a 3D projection using coordinates (x, y, α) : the two null lines in the (x, α) -plane extrude along the y -direction into the two planes

$$\alpha = x, \quad \alpha = -x.$$

These are the “zero-divisor planes” of the projected 3D scene.

Projected 3D sketch in coordinates (x, y, α)



Inside wedge: $x^2 - \alpha^2 > 0$

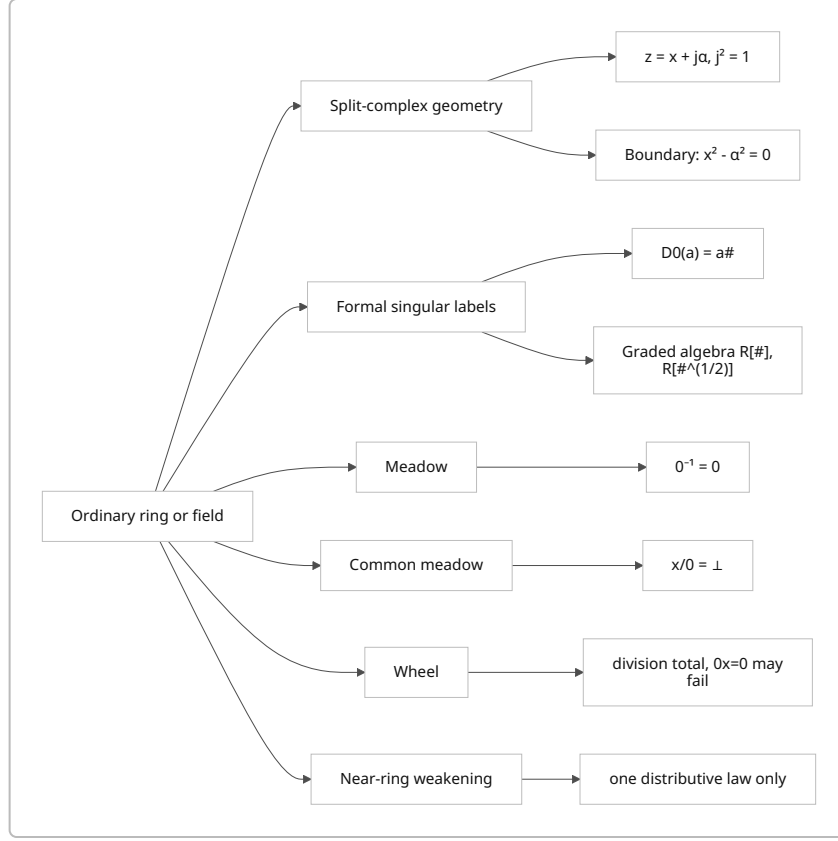
On planes: $x^2 - \alpha^2 = 0$

Outside wedge: $x^2 - \alpha^2 < 0$

Suggested caption: Split-complex singular geometry projected into (x, y, α) : the zero-divisor set $x^2 - \alpha^2 = 0$ appears as the two planes $\alpha = \pm x$ extruded along depth y .

Suggested rendering parameters: dark background, x -axis cyan, α -axis violet, null planes translucent red, $x^2 - \alpha^2 > 0$ region teal, $x^2 - \alpha^2 < 0$ region magenta, camera azimuth 30° , elevation 18° , perspective 35 mm equivalent.

The flow of algebraic interpretations can be summarized as follows.



The move from split-complex geometry to the 5D metric is conceptually straightforward but mathematically more consequential. Start from

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 + d\alpha^2.$$

Let

$$v^2 = v_x^2 + v_y^2 + v_z^2, \quad u = \frac{d\alpha}{dt}.$$

Then along a massive worldline, using $ds^2 = c^2 d\tau^2$,

$$c^2 d\tau^2 = c^2 dt^2 - v^2 dt^2 + u^2 dt^2.$$

Divide by dt^2 :

$$c^2 \left(\frac{d\tau}{dt} \right)^2 = c^2 - v^2 + u^2.$$

Hence

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2 - u^2}{c^2}}, \quad \gamma_{5D} = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v^2 - u^2}{c^2}}}.$$

This derivation is formal and correct. Its singular boundary is

$$\Delta_{5D} = 1 - \frac{v^2 - u^2}{c^2} = 0,$$

equivalently

$$v^2 - u^2 = c^2.$$

This is the exact analogue of the split-complex null condition. The two structures line up neatly:

$$x^2 - \alpha^2 = 0 \quad \longleftrightarrow \quad v^2 - u^2 = c^2.$$

However, the interpretation matters. In split-complex algebra, the null set is a zero-divisor boundary in a perfectly respectable commutative algebra. In the 5D metric, the corresponding null set lives in a spacetime with **two timelike signs**. That is a much stronger step.

This is where the critique becomes sharp. Standard Lorentzian geometry assumes one timelike direction, i.e. a metric of Lorentzian signature such as $(-, +, \dots, +)$, with a causal cone that splits into two connected components, future and past. Minguzzi's causal review explicitly formulates Lorentzian geometry in that setting. By contrast, the metric above has signature $(+, -, -, +)$, or equivalently index 2. It is therefore **not Lorentzian** in the usual sense; it is pseudo-Riemannian with two time directions. ¹⁴

That distinction has an immediate geometric consequence. In one-time Minkowski space, the timelike set is a **double cone** with two connected components. In a two-time metric, the timelike set is connected. One can see this already in the pure (t_1, t_2) -plane: the condition

$$t_1^2 + t_2^2 > 0$$

defines $\mathbb{R}^2 \setminus \{0\}$, which is connected. So there is no natural future/past separation analogous to ordinary SR. This is not merely an aesthetic difference; it is the mathematical seed of the causality difficulties discussed in the physics literature.

Indeed, one can construct a closed timelike polygon in flat two-time geometry. Consider three vectors in the (t_1, t_2) -plane:

$$v_1 = (1, 0), \quad v_2 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad v_3 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Each has positive norm $t_1^2 + t_2^2 = 1$, so each is timelike with respect to the $++$ form, and

$$v_1 + v_2 + v_3 = 0.$$

Therefore these three timelike segments form a closed loop. This is a self-contained demonstration that ordinary causal intuition is not preserved by simply adding a second timelike coordinate.

Physical plausibility and the status of the alpha axis

The physically strongest criticism is that the α -axis does not merely extend SR; it **changes its axioms**. In standard special relativity one works on Minkowski spacetime with one timelike direction; proper time and

causal structure are defined in that setting. The Cambridge GR notes quoted above derive proper time from exactly that one-time signature. Once the sign $+d\alpha^2$ is inserted into a mostly-minus convention, α is not spacelike; it is a second timelike direction. ¹⁵

That point is often missed, so it is worth stating carefully. If the modeler wants α to be a **spatial** extra dimension, the metric must be

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 - d\alpha^2.$$

Then the corresponding modified dilation factor would be

$$\gamma_{\text{space-like } \alpha} = \frac{1}{\sqrt{1 - \frac{v^2 + u^2}{c^2}}},$$

which makes the singular boundary stricter, not looser:

$$v^2 + u^2 = c^2.$$

So a spacelike α does **not** support the intended “bypass the $v = c$ singularity” narrative. Only a timelike α does.

The literature on time-like extra dimensions makes exactly this tradeoff explicit. Quiros’ review compares timelike and spacelike extra dimensions and discusses causality and unitarity issues; Bars’ two-time physics treats two-time theories as possible only because a nontrivial gauge symmetry removes unphysical degrees of freedom. Bars is explicit that more than one timelike direction produces ghost problems unless the extra times are controlled by gauge constraints, and that more than two times are already too many for the $Sp(2, \mathbb{R})$ mechanism to cure. ¹⁶

This yields a balanced defense of the alpha-axis idea.

The defense is mathematical: if the goal is to create a **computational or geometric regularization layer** around a singular boundary, then introducing α as an auxiliary coordinate is reasonable. It behaves much like embedding a difficult boundary into a higher-dimensional state space where it can be visualized and stratified.

The criticism is physical: if the goal is to claim a new physical spacetime in which ordinary light-speed barriers are avoided, then one must answer at least four unresolved questions. First, what is the operational meaning of α ? Second, why has this extra timelike direction not appeared in experiments? Third, how are causality and unitarity preserved? Fourth, why does the model not require the kind of gauge structure that Bars’ two-time theories require? On current evidence, those questions are not answered by the bare metric ansatz alone. ¹⁷

A second physical criticism concerns observables. If $u = d\alpha/dt$ is a physical velocity component, then the modified time-dilation law changes measurable kinematics. If it is not physical but gauge-like, then the model is closer to an auxiliary-variable formalism than to new spacetime physics. Both readings are possible, but they are not equivalent. They should not be mixed rhetorically.

The most defensible wording is therefore:

“The alpha-axis is a modeling coordinate for singular structure unless and until a physical dynamics for it is specified”

That sentence is rigorous, modest, and compatible with both mathematical and physical caution.

Simulation semantics, explicit examples, and recommendations

For simulation, the central question is not “what is $1/0$?” but “what state should the program enter when a denominator crosses a singular boundary?” The right answer is to use a **typed state machine**, not a fake number.

A good minimal value type is:

Real(r), **Singular**(c, λ, b), **Bottom**.

Here c is the coefficient, λ is the singular order, and b is an optional branch label. For example:

$$\frac{a}{0} \rightsquigarrow \mathbf{Singular}(a, 1, \text{plain}), \quad \frac{a}{\sqrt{0}} \rightsquigarrow \mathbf{Singular}(a, \frac{1}{2}, \text{plain}).$$

For split-complex boundary tracking one can also attach a branch

$$b \in \{+, -\}$$

to indicate which null sheet or approach direction was hit.

A numerically robust transition rule is:

```

flowchart TD
    A[Input expression] --> B[Compute invariant Δ]
    B --> C{ |Δ| > ε ? }
    C -- Yes --> D[Real domain]
    C -- No --> E[Singular boundary]
    E --> F{ numerator also ~ 0 ? }
    F -- Yes --> G[Bottom or indeterminate]
    F -- No --> H[Create Singular tag]
    H --> I[Propagate symbolically or quarantine]
    I --> J[Continue numeric evaluation]

```

For the split-complex geometry, the invariant is

$$\Delta_{\text{split}} = x^2 - \alpha^2.$$

For the 5D metric, it is

$$\Delta_{5D} = 1 - \frac{v^2 - u^2}{c^2}.$$

Then the simulation states can be:

$$\begin{aligned}\text{REAL_DOMAIN} &\iff \Delta > \varepsilon, \\ \text{SINGULAR_BOUNDARY} &\iff |\Delta| \leq \varepsilon, \\ \text{ALPHA_DOMAIN} &\iff \Delta < -\varepsilon.\end{aligned}$$

The core arithmetic rules for a symbolic engine can be:

$$\begin{aligned}\text{Real}(a) + \text{Real}(b) &= \text{Real}(a + b), \\ \text{Singular}(a, \lambda, b) + \text{Singular}(c, \lambda, b) &= \text{Singular}(a + c, \lambda, b), \\ \text{Singular}(a, \lambda, b) \cdot \text{Singular}(c, \mu, d) &= \text{Singular}(ac, \lambda + \mu, \text{merge}(b, d)), \\ \text{Real}(a) \cdot \text{Singular}(c, \lambda, b) &= \text{Singular}(ac, \lambda, b),\end{aligned}$$

and any structurally indeterminate case, such as mixing incompatible branch metadata or encountering $\frac{0}{0}$, becomes **Bottom**.

This is exactly the computational form of the graded singular algebra:

$$(a\sigma^\lambda)(b\sigma^\mu) = ab\sigma^{\lambda+\mu}.$$

The visualization layer is then simple. Use x, y, z as geometric coordinates and map singular metadata to appearance. A good encoding is:

$$\text{hue} \leftrightarrow \text{sign}(\Delta), \quad \text{brightness} \leftrightarrow -\log_{10} |\Delta|, \quad \text{size} \leftrightarrow \lambda, \quad \text{shape} \leftrightarrow b.$$

For example, the split-complex null planes $\alpha = \pm x$ can be red translucent sheets; points entering the singular boundary can pulse and then be re-rendered as violet points placed on or near the α -axis with a label $a\#$.

Small explicit examples show the difference among the models.

In the split-complex algebra,

$$z = 2 + j \quad \Rightarrow \quad N(z) = 4 - 1 = 3, \quad z^{-1} = \frac{2 - j}{3}.$$

But

$$z = 1 + j \quad \Rightarrow \quad N(z) = 0,$$

so z is not invertible. This is a genuine geometric singular boundary, not a division-by-zero number. 18

In the formal singular algebra,

$$D_0(2) + D_0(3) = 2\sigma + 3\sigma = 5\sigma,$$

$$D_0(2) D_0(3) = 6\sigma^2.$$

No contradiction appears because σ was never declared to satisfy $0\sigma = 1$.

In the near-ring toy model $(G, +, \star)$ with $a \star b = b$,

$$0 \star K = K.$$

So if the additive group contains a nonzero element named 1, then

$$0 \star 1 = 1.$$

This proves that such behavior is algebraically possible once ring laws are weakened, but it also shows how far one has moved away from ordinary arithmetic.

In the 5D kinematics,

$$v = 1.2c, \quad u = 0.8c$$

gives

$$\Delta_{5D} = 1 - (1.2^2 - 0.8^2) = 1 - (1.44 - 0.64) = 0.20,$$

so

$$\gamma_{5D} = \frac{1}{\sqrt{0.20}} \approx 2.236.$$

Thus a superluminal 3-speed becomes compatible with a real dilation factor **only because** an extra timelike degree of freedom has been postulated. That is a feature of the altered model, not a theorem of SR.

The table below gives the practical recommendation.

The recommendations synthesize the formal analysis above with the literature-backed properties of split-complex algebras, wheels, meadows, common meadows, near-rings, and higher-signature metrics. ¹⁹

Goal	Best formalism	Why
Keep numerator information for $a/0$	Formal singular operator $D_0(a) = a\#$	Clean, consistent, simulation-friendly
Represent singularity as geometric boundary	Split-complex plane $x + j\alpha$	Zero-divisor locus is explicit and visual
Totalize arithmetic with rigorous literature support	Common meadow	$x/0 = \perp$ is clean and logically disciplined
Explore full “division always defined” algebra	Wheel	Most ambitious established algebraic alternative

Goal	Best formalism	Why
Demonstrate how zero-absorption can fail	Near-ring	Good counterexample environment
Preserve standard SR physics	Do not use $+d\alpha^2$	Two-time signature changes causality
Prototype visualization in code	Tagged state machine + split-complex geometry	Best balance of rigor and usability

The bottom-line recommendation is therefore precise.

For **mathematical theory**, the cleanest statement of the proposal is:

Use $D_0(a) = a\#$ as a formal singular-label operator, not as $a \cdot 0^{-1}$.

For **geometry**, the cleanest companion model is:

$z = x + j\alpha, \quad j^2 = 1, \quad \text{singular boundary } x^2 - \alpha^2 = 0.$

For **physics**, the honest statement is:

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 + d\alpha^2$$

defines a speculative pseudo-Riemannian model with two timelike directions, not a conservative extension of special relativity.

And for **simulation**, the correct operational rule is:

$\Delta \approx 0 \Rightarrow$ create a tagged singular state, do not coerce to ∞ .

That is the rigorous version of the user's core intuition: not "division by zero has become an ordinary number," but rather "division by zero has become a typed singular object and, optionally, a geometric boundary that computation can preserve, visualize, and quarantine instead of discarding."

1 4 9 13 18 19 The Split Complex Numbers

<https://books.physics.oregonstate.edu/GELG/csplit.html>

2 8 15 damtp.cam.ac.uk

<https://www.damtp.cam.ac.uk/user/us248/Lectures/Notes/grII.pdf>

3 5 2001-11.dvi

<https://www2.math.su.se/reports/2001/11/2001-11.pdf>

6 12 [0901.0823] Meadows and the equational specification of division

<https://arxiv.org/abs/0901.0823>

7 11 **On distributively generated near-rings**

<https://www.cambridge.org/core/services/aop-cambridge-core/content/view/D436F9D0CFD60F7DA1405042EB60D648/S001309150001275Xa.pdf/div-class-title-on-distributively-generated-near-rings-a-href-fn01-ref-type-fn-span-class-sup-1-span-a-div.pdf>

10 [2307.04270] **A Complete Finite Axiomatisation of the Equational Theory of Common Meadows**

<https://arxiv.org/abs/2307.04270>

14 **Lorentzian causality theory**

<https://inspirehep.net/files/7a575d682fd15aaad8463a778fac1609>

16 [0707.0714] **Time-like vs Space-like Extra Dimensions**

<https://arxiv.org/abs/0707.0714>

17 **arxiv.org**

<https://arxiv.org/pdf/0804.1585>